Linear Algebra: Midterm Study Guide

Vectors · Matrices · Elimination · Inverses · LU

1. Vectors

1.1 What a vector is, and what you can do with it

Key idea

A **vector** is an ordered list of numbers. Write it as a column $\begin{bmatrix}3\\4\end{bmatrix}$ or in a line as $(3,4)$. Read it as a trip: " go 3 steps right and 4 steps up". The order matters: $(3,4) \neq (4,3)$. The numbers are called the **components**.

All operations work **component by component** (spot by spot).

Example

Add: $(1,2)+(3,1)=(1+3,2+1)=(4,3)$
Subtract: $(5,5)-(2,1)=(3,4)$
Multiply by a number (scalar): $2(3,4)=(6,8)$, every component is multiplied
Negative: $-1(3,4)=(-3,-4)$. The vector keeps its length but points the opposite way.

Method: the trip from point Q to point P

Always **END minus START**: the vector from Q to P is $P-Q$. Example: $Q=(1,1)$ and $P=(4,5)$, so the trip is $P-Q=(4-1,5-1)=(3,4)$.

Watch out

Vectors can only be added if they have the **same number of components**. You cannot add $(1,2)$ and $(1,2,3)$.

1.2 Linear combinations (mixing vectors)

Key idea

A **linear combination** of $\boldsymbol{v}$ and $\boldsymbol{w}$ is $c\boldsymbol{v}+d\boldsymbol{w}$, where c and d are any numbers. You take some amount of each vector and add the results.

Example

$2\begin{bmatrix}1\\0\end{bmatrix}+3\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}2\\3\end{bmatrix}$
$0.75\begin{bmatrix}4\\0\end{bmatrix}+1.5\begin{bmatrix}2\\2\end{bmatrix}=\begin{bmatrix}3\\0\end{bmatrix}+\begin{bmatrix}3\\3\end{bmatrix}=\begin{bmatrix}6\\3\end{bmatrix}$

The set of **all** combinations you can build tells you the shape you can reach:

You have	All combinations fill
One nonzero vector	A line through the origin
Two vectors pointing in different directions	A flat sheet (a plane)
Two vectors on the same line	Still only a line
Three vectors in 3D that do not lie in one plane	All of 3D space

Special cases: same line? same plane?

1. **Same line (2 vectors):** is one a multiple of the other? $(1,2)$ and $(2,4)$: yes, $(2,4)=2(1,2)$. $(1,2)$ and $(3,4)$: no number works, so they point in different directions.
2. **Same plane (3 vectors):** if some combination (not all zero) gives the zero vector, they share a plane. For example $(1,2,3)+(-3,1,-2)+(2,-3,-1)=(0,0,0)$, so they lie in one plane.
3. **The zero vector** $(0,0)$ is a combination of anything (take zero of each). It does not add a new direction.

Classic exam examples

Clock: the 12 hour arrows from the centre add up to zero because opposite arrows cancel. If you remove the 2 o'clock arrow, the remaining 11 add up to the arrow opposite to it, the 8 o'clock arrow.

Two unknown vectors: $\boldsymbol{v}+\boldsymbol{w}=(5,1)$ and $\boldsymbol{v}-\boldsymbol{w}=(1,5)$. Add the equations: $2\boldsymbol{v}=(6,6)$, so $\boldsymbol{v}=(3,3)$. Then $\boldsymbol{w}=(5,1)-(3,3)=(2,-2)$.

Cube: the centre of the unit cube is $(0.5,0.5,0.5)$ and it has 12 edges. All mixes of $(1,0,0)$ and $(1,1,0)$ fill the floor, the plane $z=0$.

1.3 Dot products, lengths, angles

Definition

Multiply matching components, then add everything:

$$(a_1,a_2,a_3)\cdot(b_1,b_2,b_3)=a_1b_1+a_2b_2+a_3b_3$$

The answer is a **single number**, not a vector.

Example

$(1,2,3)\cdot(4,-5,6)=1\cdot4+2\cdot(-5)+3\cdot6=4-10+18=12$

Length of a vector

The length is the square root of the vector dotted with itself:

$$\|\boldsymbol{v}\|=\sqrt{\boldsymbol{v}\cdot\boldsymbol{v}}$$

For $\boldsymbol{v}=(3,4)$: $\|\boldsymbol{v}\|=\sqrt{3^2+4^2}=\sqrt{9+16}=\sqrt{25}=5$.

Unit vector (a vector of length 1)

Divide the vector by its own length: $\boldsymbol{u}=\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}$. For $(3,4)$ it is $(\frac{3}{5},\frac{4}{5})$. Check: $\sqrt{\left(\frac{3}{5}\right)^2+\left(\frac{4}{5}\right)^2}=\sqrt{\frac{9}{25}+\frac{16}{25}}=\sqrt{1}=1$.

Perpendicular (orthogonal) vectors

Two vectors are perpendicular **exactly when their dot product is 0**. Example: $(-4,3)\cdot(3,4)=-12+12=0$, so they are perpendicular.

Angle between two vectors

$$\cos\theta=\frac{\boldsymbol{v}\cdot\boldsymbol{w}}{\|\boldsymbol{v}\|\|\boldsymbol{w}\|}$$

For $(3,4)$ and $(4,3)$: dot $=24$, both lengths are 5, so $\cos\theta=\frac{24}{25}$. Reading the sign of the dot product: positive means an angle below 90°, zero means exactly 90°, negative means more than 90°.

Normal vector of a line

The line $2x+4y=0$ can be written as $(2,4)\cdot(x,y)=0$, so $(2,4)$ is perpendicular to every point on the line. It is the **normal vector**. The parallel line $2x+4y=8$ has the **same** normal $(2,4)$. Only the right-hand side changes, which shifts the line.

Watch out

- You cannot make a unit vector from the zero vector (its length is 0 and you cannot divide by 0).
- Perpendicular to $(3,4)$: there are only **2 directions** ($(-4,3)$ and $(4,-3)$), but **infinitely many vectors** if any length is allowed. Read the question: "how many directions" or "how many vectors"?
- $\cos\theta$ is always between -1 and 1. If you get $\frac{24}{25}$, fine; if you get 1.3, you made an arithmetic mistake.

2. Matrices and the equation $Ax=b$

2.1 Matrices

Key idea

A **matrix** is a rectangular grid of numbers. An $m\times n$ matrix has m rows and n columns (rows first!). A 2×3 matrix has 2 rows and 3 columns. Two matrices can be added only if they have the **same size**, and then you add spot by spot.

Special matrices you must recognise:

- **Identity** I: ones on the diagonal, zeros elsewhere. It changes nothing: $\begin{bmatrix}1&0\\0&1\end{bmatrix}\begin{bmatrix}7\\8\end{bmatrix}=\begin{bmatrix}7\\8\end{bmatrix}$.
- **Zero matrix:** all entries are 0.
- **Diagonal matrix:** nonzero numbers only on the main diagonal.
- **Triangular matrix:** all zeros above the diagonal (lower triangular) or all zeros below it (upper triangular).

2.2 Matrix times vector: two ways, same answer

$$\begin{bmatrix}1&2\\3&4\end{bmatrix}\begin{bmatrix}5\\6\end{bmatrix}=\begin{bmatrix}17\\39\end{bmatrix}$$

Rows way

Take the dot product of **each row** with the vector. Row 1: $1\cdot5+2\cdot6=17$. Row 2: $3\cdot5+4\cdot6=39$.

Columns way (the most important view)

The answer is a **mix of the columns**, using the entries of the vector as amounts:

$$5\begin{bmatrix}1\\3\end{bmatrix}+6\begin{bmatrix}2\\4\end{bmatrix}=\begin{bmatrix}5\\15\end{bmatrix}+\begin{bmatrix}12\\24\end{bmatrix}=\begin{bmatrix}17\\39\end{bmatrix}$$

So $A\boldsymbol{x}$ is always "a combination of the columns of A".

Try it

$\begin{bmatrix}0&2&-1\\-2&1&1\end{bmatrix}\begin{bmatrix}2\\-1\\-1\end{bmatrix}=\begin{bmatrix}0\cdot2+2\cdot(-1)+(-1)(-1)\\-2\cdot2+1\cdot(-1)+1\cdot(-1)\end{bmatrix}=\begin{bmatrix}-1\\-6\end{bmatrix}$
Rotate by 90°: (x,y) becomes $(-y,x)$. **Rotate by 180°:** every sign flips.
Difference matrix: (x_1,x_2,x_3) becomes (x_1,x_2-x_1,x_3-x_2), so $(1,3,6)$ becomes $(1,2,3)$. The **sum matrix** undoes it with running totals: $(1,2,3)$ becomes $(1,1+2,1+2+3)=(1,3,6)$.

2.3 Two ways to see the same equations

Take the system $x-2y=1$ and $3x+2y=11$. The solution is $x=3$, $y=1$.

Row picture

Each equation is a **line**. The solution is the point where the lines cross: $(3,1)$. With 3 unknowns, each equation is a **plane** instead.

Column picture

Write it as $A\boldsymbol{x}=\boldsymbol{b}$ and ask: "how much of each column makes $\boldsymbol{b}$?"

$$3\begin{bmatrix}1\\3\end{bmatrix}+1\begin{bmatrix}-2\\2\end{bmatrix}=\begin{bmatrix}3-2\\9+2\end{bmatrix}=\begin{bmatrix}1\\11\end{bmatrix}$$

Columns are the ingredients, $\boldsymbol{b}$ is the dish, and solving means finding the amounts.

2.4 How many solutions can a system have?

Key idea

Only three cases exist: **exactly one** solution, **no** solution, or **infinitely many**. **Never exactly two.** If two different solutions existed, every point on the line between them would also be a solution.

In two dimensions: lines that **cross** give exactly one solution, **parallel** lines give none, and two copies of the **same line** give infinitely many.

Infinitely many solutions

$x+y+z=2,\quad x+2y+z=3,\quad 2x+3y+2z=5.$
The third equation is the first plus the second $(2+3=5)$, so it gives **no new information**. Subtracting eq. 1 from eq. 2 gives $y=1$, then $x+z=1$. Solutions: $(1,1,0)$, $(0,1,1)$, $(2,1,-1)$, and so on. In general $(1-t,1,t)$ for any number t.

No solution

Change the last equation to $2x+3y+2z=9$. Now eq. 1 + eq. 2 − eq. 3 gives $0=2+3-9=-4$. This is impossible, so there is **no solution**.

2.5 Dependent columns and singular matrices

Key idea

A column is **dependent** if it is a mix of the other columns. Equivalently: some combination of the columns, with not all amounts zero, gives the zero vector. A square matrix with dependent columns is called **singular**, and it has no inverse.

Example

Columns $(1,2,3)$, $(4,5,6)$, $(7,8,9)$:
$1\cdot(1,2,3)-2\cdot(4,5,6)+1\cdot(7,8,9)=(1-8+7,2-10+8,3-12+9)=(0,0,0)$. Dependent.

Method: find c so that the columns are dependent

Columns $(1,1,0)$, $(3,2,1)$, $(7,4,c)$. Try to write the third as a mix of the first two: $-2(1,1,0)+3(3,2,1)=(-2+9,-2+6,0+3)=(7,4,3)$. For the third column to be that mix, $c=3$.

Shortcut to prove there is no solution

Rows $(1,3,5)$, $(1,2,-3)$, $(2,5,2)$ with right-hand sides 4, 5, 8. Combine: row 1 + row 2 − row 3 gives $0=0=0$ on the left. The right side gives $4+5-8=1$. So $0=1$: **no solution**. Rule: if rows combine to zero on the left but not on the right, the system is impossible.

3. Elimination: solving systems step by step

3.1 The idea

Key idea

Elimination turns a hard system into an easy one **without changing the solution**. The goal of the first phase (forward elimination) is a **staircase**: zeros **below** each pivot. Then the last equation has one unknown, the one above it has two, and you climb back up.

Allowed moves (these never change the solution):
1. Subtract a multiple of one row from another row.
2. Swap two rows.
3. Multiply a row by a nonzero number.

Always do the **same move to the right-hand side** (the numbers after the equals sign). The easiest way is to write the **augmented matrix** $[A\mid\boldsymbol{b}]$ and treat each row as one equation.

3.2 Method: forward elimination (zeros below the pivots)

Step by step

1. Look at the **first column**. The entry in the top row is the first **pivot**. It must not be 0.
2. For every row below, find the **multiplier** $\ell=\frac{\text{number to kill}}{\text{pivot}}$.
3. Replace that row by: **row − $\ell\cdot$ (pivot row)**. The entry below the pivot becomes **0**. The rest of the row changes too, as does the right-hand side.
4. Move one column right and one row down. The new entry on the diagonal is the **second pivot**. Repeat steps 2 and 3 to get zeros below it.
5. Continue until the matrix is **upper triangular** (all zeros below the diagonal).

Back substitution

Start from the bottom row (one unknown), solve it, put its value into the row above, and so on up to the top.

Small example

Solve $x-2y=1$, $3x+2y=11$.

$$\left[\begin{array}{cc|c}1&-2&1\\3&2&11\end{array}\right]$$

Pivot 1 is the 1 in the top-left. To kill the 3 below it, the multiplier is $\frac{3}{1}=3$. Do R_2-3R_1:

$$\left[\begin{array}{cc|c}1&-2&1\\3&2&11\end{array}\right]\xrightarrow{R_2-3R_1}\left[\begin{array}{cc|c}1&-2&1\\0&8&8\end{array}\right]$$

Pivot 2 is the 8. The matrix is now upper triangular. Back substitution: $8y=8$ gives $y=1$. Then $x-2(1)=1$ gives $x=3$.

Full 3 by 3 example

$$2x+y+z=5,\quad 4x-6y=-2,\quad -2x+7y+2z=9$$

$$\left[\begin{array}{ccc|c}2&1&1&5\\4&-6&0&-2\\-2&7&2&9\end{array}\right]$$

Column 1, pivot = 2.
Row 2 has 4: multiplier $\frac{4}{2}=2$. Do R_2-2R_1: $(4,-6,0\mid-2)-2(2,1,1\mid5)=(0,-8,-2\mid-12)$.
Row 3 has −2: multiplier $-\frac{2}{2}=-1$. Do $R_3-(-1)R_1=R_3+R_1$: $(-2,7,2\mid9)+(2,1,1\mid5)=(0,8,3\mid14)$.

$$\left[\begin{array}{ccc|c}2&1&1&5\\0&-8&-2&-12\\0&8&3&14\end{array}\right]$$

Column 2, pivot = −8 (a negative pivot is fine; only 0 is a problem).
Row 3 has 8: multiplier $\frac{8}{-8}=-1$. Do R_3+R_2: $(0,8,3\mid14)+(0,-8,-2\mid-12)=(0,0,1\mid2)$.

$$\left[\begin{array}{ccc|c}2&1&1&5\\0&-8&-2&-12\\0&0&1&2\end{array}\right]$$

The pivots are 2, −8, 1. **Back substitution:** the last row says $z=2$. Row 2: $-8y-2(2)=-12$, so $-8y=-8$ and $y=1$. Row 1: $2x+1+2=5$, so $x=1$.

Answer: $(x,y,z)=(1,1,2)$. Always check by putting it into the original equations: $4(1)-6(1)=-2$. Correct.

3.3 Method: Gauss-Jordan (zeros above the pivots and 1s on the diagonal)

Forward elimination stops at the staircase. If you want to **read the answer directly** (or find an inverse), keep going.

Step by step, starting from the upper triangular form

1. Start at the **last pivot**. Use it to make **zeros above it** (subtract multiples of the bottom row from the rows above).
2. Move up one pivot and make zeros above that one too.
3. When the matrix is **diagonal** (only the diagonal is nonzero), **divide every row by its pivot** so that each diagonal entry becomes **1**.
4. Now the left side is the identity matrix I, and the right-hand column **is** the solution.

Continuing the 3 by 3 example

Start from $\left[\begin{array}{ccc|c}2&1&1&5\\0&-8&-2&-12\\0&0&1&2\end{array}\right]$.
Zeros above the last pivot (the 1 in row 3):
R_2+2R_3 gives $(0,-8,0\mid-8)$.
R_1-1R_3 gives $(2,1,0\mid3)$.

$$\left[\begin{array}{ccc|c}2&1&0&3\\0&-8&0&-8\\0&0&1&2\end{array}\right]$$

Zero above the middle pivot (the −8): $R_1+\frac{1}{8}R_2$ gives $(2,0,0\mid3-1)=(2,0,0\mid2)$.

$$\left[\begin{array}{ccc|c}2&0&0&2\\0&-8&0&-8\\0&0&1&2\end{array}\right]$$

Make the diagonal 1s: $\frac{R_1}{2}$, $\frac{R_2}{-8}$, $\frac{R_3}{1}$:

$$\left[\begin{array}{ccc|c}1&0&0&1\\0&1&0&1\\0&0&1&2\end{array}\right]$$

The left side is I. Reading the right column: $x=1$, $y=1$, $z=2$. Same answer as before.

3.4 Special cases (what to do when things go wrong)

Case 1: a zero pivot, but there is a nonzero entry below it, so swap rows

$$\left[\begin{array}{cc|c}0&2&4\\3&4&11\end{array}\right]$$

The top-left entry is 0, so you cannot use it to kill anything. **Swap** row 1 and row 2:

$$\left[\begin{array}{cc|c}3&4&11\\0&2&4\end{array}\right]$$

It is already upper triangular. Back substitution: $2y=4$ gives $y=2$; $3x+8=11$ gives $x=1$. This is a **temporary** failure and the matrix is still fine (invertible).

Case 2: a zero pivot and nothing nonzero below it, so a pivot is missing (singular)

If the pivot position is 0 **and every entry below it is also 0**, there is no way to fix it by swapping. That column has **no pivot**. The matrix is **singular**, and one of two things happens:
1. **No solution**, if you reach a row like $(0,0,0\mid4)$, which says $0=4$. This is a contradiction.
2. **Infinitely many solutions**, if you reach $(0,0,0\mid0)$, which says $0=0$. That row is true and useless.

The system is singular in both cases. The right-hand side decides which one you get.

Case 2a: no solution, full elimination

$x+y+z=2,\quad x+2y+z=3,\quad 2x+3y+2z=9.$

$$\left[\begin{array}{ccc|c}1&1&1&2\\1&2&1&3\\2&3&2&9\end{array}\right]\xrightarrow{R_2-R_1,\,R_3-2R_1}\left[\begin{array}{ccc|c}1&1&1&2\\0&1&0&1\\0&1&0&5\end{array}\right]\xrightarrow{R_3-R_2}\left[\begin{array}{ccc|c}1&1&1&2\\0&1&0&1\\0&0&0&4\end{array}\right]$$

Column 3 has no pivot, and the last row says $0=4$. **No solution.**

Case 2b: infinitely many solutions, and how to write them

Same left side, but the last right-hand side is 5:

$$\left[\begin{array}{ccc|c}1&1&1&2\\1&2&1&3\\2&3&2&5\end{array}\right]\longrightarrow\left[\begin{array}{ccc|c}1&1&1&2\\0&1&0&1\\0&0&0&0\end{array}\right]$$

The last row is $0=0$. The pivot columns are columns 1 and 2. Column 3 has no pivot, so z is a **free variable**: you choose it (call it t) and the other unknowns follow. Row 2: $y=1$. Row 1: $x+y+z=2$ gives $x=1-t$.

$$(x,y,z)=(1-t,1,t)\quad\text{for any number } t$$

Check with $t=0$: $(1,1,0)$. With $t=1$: $(0,1,1)$. With $t=-1$: $(2,1,-1)$.

Common mistakes

- The multiplier uses the **pivot as the denominator**, never the other way round.
- Subtracting a row means **all** entries of the row, including the right-hand side.
- A pivot may be negative or a fraction. Only 0 is forbidden.
- Number of pivots = number of unknowns means **exactly one solution** (for a square system, invertible).
- Fewer pivots than unknowns means singular: then $0=0$ rows give infinitely many solutions, and $0=$ nonzero rows give none.

3.5 Elimination written with matrices

Key idea

Each row operation is a multiplication on the left by an **elimination matrix** E. The matrix E_{ij} is the identity matrix with one extra entry $-\ell_{ij}$ in row i, column j. It means "subtract ℓ_{ij} times row j from row i".

Example

$$E_{21}=\begin{bmatrix}1&0&0\\-2&1&0\\0&0&1\end{bmatrix}$$

subtracts $2\times$ row 1 from row 2. Its **inverse** adds the $2\times$ row 1 back: $E_{21}^{-1}=\begin{bmatrix}1&0&0\\2&1&0\\0&0&1\end{bmatrix}$.

For the 3 by 3 example, with $A=\begin{bmatrix}2&1&1\\4&-6&0\\-2&7&2\end{bmatrix}$:

$$E_{21}=\begin{bmatrix}1&0&0\\-2&1&0\\0&0&1\end{bmatrix},\quad E_{31}=\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix},\quad E_{32}=\begin{bmatrix}1&0&0\\0&1&0\\0&1&1\end{bmatrix}$$

and $E_{32}E_{31}E_{21}A=U$, where $U=\begin{bmatrix}2&1&1\\0&-8&-2\\0&0&1\end{bmatrix}$. Apply them in order: first E_{21}, then E_{31}, then E_{32}.

Row swap: the permutation matrix

$$P=\begin{bmatrix}0&1\\1&0\end{bmatrix},\quad P\begin{bmatrix}1&2\\3&4\end{bmatrix}=\begin{bmatrix}3&4\\1&2\end{bmatrix}$$

Multiplying by P on the left **swaps the rows**.

4. Matrix multiplication, inverses and LU

4.1 Rules for matrix operations

How to multiply

Each entry of AB is **(row of A) · (column of B)**. The **size rule**: $(m\times n)(n\times p)$ gives $m\times p$. The two inner numbers must match. A (2×3) times a (3×2) gives 2×2.

Regrouping is allowed: $(AB)C=A(BC)$ and $A(B+C)=AB+AC$.

Order matters! In general AB is not BA

$$\begin{bmatrix}1&2\\3&4\end{bmatrix}\begin{bmatrix}5&6\\7&8\end{bmatrix}=\begin{bmatrix}19&22\\43&50\end{bmatrix}\quad\text{but}\quad\begin{bmatrix}5&6\\7&8\end{bmatrix}\begin{bmatrix}1&2\\3&4\end{bmatrix}=\begin{bmatrix}23&34\\31&46\end{bmatrix}$$

So you cannot swap the order, and you cannot cancel A from $AB=AC$ unless A is invertible.

Column times row gives a full grid

$$\begin{bmatrix}1\\2\\3\end{bmatrix}\begin{bmatrix}4&5&6\end{bmatrix}=\begin{bmatrix}4&5&6\\8&10&12\\12&15&18\end{bmatrix}$$

(a 3×1 times a 1×3 gives 3×3). Every row is a multiple of $(4,5,6)$.

4.2 Inverse matrices

Key idea

The inverse A^{-1} undoes A: $A^{-1}A=I$ and $AA^{-1}=I$. If $A\boldsymbol{x}=\boldsymbol{b}$ and A has an inverse, then $\boldsymbol{x}=A^{-1}\boldsymbol{b}$. Only square matrices with **a full set of pivots** have an inverse.

The 2 by 2 formula

$$\begin{bmatrix}a&b\\c&d\end{bmatrix}^{-1}=\frac{1}{ad-bc}\begin{bmatrix}d&-b\\-c&a\end{bmatrix}$$

Swap the two diagonal entries, flip the signs of the other two, and divide by $ad-bc$. If $ad-bc=0$ there is **no inverse**.

Example

$\begin{bmatrix}1&3\\2&7\end{bmatrix}$ has $ad-bc=1\cdot7-3\cdot2=1$, so $\begin{bmatrix}1&3\\2&7\end{bmatrix}^{-1}=\begin{bmatrix}7&-3\\-2&1\end{bmatrix}$.
Check: $\begin{bmatrix}1&3\\2&7\end{bmatrix}\begin{bmatrix}7&-3\\-2&1\end{bmatrix}=\begin{bmatrix}7-6&-3+3\\14-14&-6+7\end{bmatrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}$.

Gauss-Jordan method for any size

Write $[A\mid I]$ and do elimination (forward, then upward, then divide by pivots) until the left side becomes I. The right side is then A^{-1}:

$$[A\mid I]\longrightarrow[I\mid A^{-1}]$$

For the matrix above: $[A\mid I]=\left[\begin{array}{cc|cc}1&3&1&0\\2&7&0&1\end{array}\right]$. Do R_2-2R_1 to get $\left[\begin{array}{cc|cc}1&3&1&0\\0&1&-2&1\end{array}\right]$. Then R_1-3R_2 to get $\left[\begin{array}{cc|cc}1&0&7&-3\\0&1&-2&1\end{array}\right]$. The right half is A^{-1}.

When there is no inverse

A matrix has **no inverse** if some **nonzero** $\boldsymbol{x}$ has $A\boldsymbol{x}=\boldsymbol{0}$ (the columns are dependent). Example:

$$\begin{bmatrix}1&2\\2&4\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

and indeed $ad-bc=4-4=0$. In Gauss-Jordan you would hit a row of zeros on the left: a missing pivot.

Two more rules

Product: $(AB)^{-1}=B^{-1}A^{-1}$. The order **reverses**, like taking off shoes and socks.
Diagonal matrix: invert each diagonal number, so 2 becomes $\frac{1}{2}$.

4.3 $A=LU$: the elimination recipe

Key idea

Forward elimination turns A into the upper triangular matrix U (the pivots are on its diagonal). The **multipliers** you used are stored in L, a **lower triangular** matrix with **1s on the diagonal**. Together, $A=LU$. Why the multipliers go in with a plus sign: $E_{32}E_{31}E_{21}A=U$ means $A=E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U$, and inverting an E flips the sign of its entry, so the multipliers appear as they are. Put each multiplier ℓ_{ij} in row i, column j of L.

Small

$$\begin{bmatrix}1&2\\3&4\end{bmatrix}=\begin{bmatrix}1&0\\3&1\end{bmatrix}\begin{bmatrix}1&2\\0&-2\end{bmatrix}$$

Multiplier 3 (row 2 − 3 × row 1) goes into position (2,1) of L.

Our 3 by 3 system

The multipliers were $\ell_{21}=2$, $\ell_{31}=-1$, $\ell_{32}=-1$.

$$\begin{bmatrix}2&1&1\\4&-6&0\\-2&7&2\end{bmatrix}=\begin{bmatrix}1&0&0\\2&1&0\\-1&-1&1\end{bmatrix}\begin{bmatrix}2&1&1\\0&-8&-2\\0&0&1\end{bmatrix}$$

Check one entry: row 3 of L times U is $-1(2,1,1)-1(0,-8,-2)+1(0,0,1)=(-2,7,2)$, which is row 3 of A.

Solving A x = b with L and U (two easy steps)

Since $A\boldsymbol{x}=LU\boldsymbol{x}=\boldsymbol{b}$, split it in two:
1. **Forward:** solve $L\boldsymbol{c}=\boldsymbol{b}$ from the top down.
2. **Back:** solve $U\boldsymbol{x}=\boldsymbol{c}$ from the bottom up.

For $\boldsymbol{b}=(5,-2,9)$: forward gives $c_1=5$, then $2\cdot5+c_2=-2$ so $c_2=-12$, then $-5+12+c_3=9$ so $c_3=2$. So $\boldsymbol{c}=(5,-12,2)$. Back substitution on $U\boldsymbol{x}=\boldsymbol{c}$ gives $\boldsymbol{x}=(1,1,2)$. This matches the earlier answer, and $\boldsymbol{c}$ is exactly the right-hand column you got during elimination.

Special cases and cost

- If rows had to be swapped, the factorisation becomes $PA=LU$ (the swaps are collected in P).
- Elimination on A is the expensive part, about $\frac{n^3}{3}$ steps. Once L and U are saved, every **new** $\boldsymbol{b}$ costs only about n^2 steps.

5. Exam checklist

Before you hand in

1. Solving a system? Did you do the **same row operation on the right-hand side**?
2. Did you **plug your answer back** into the original equations?
3. Is every pivot nonzero? If not: swap rows, or conclude singular (no solution or infinitely many).
4. Inverse of a 2×2? Check $ad-bc\neq0$, then multiply AA^{-1} to see I.
5. Matrix products: sizes match, and the order is the order given.
6. Lengths: did you take the square root of the **whole** sum?